Scalability of the Global Optimization Solver on Trillium Cluster

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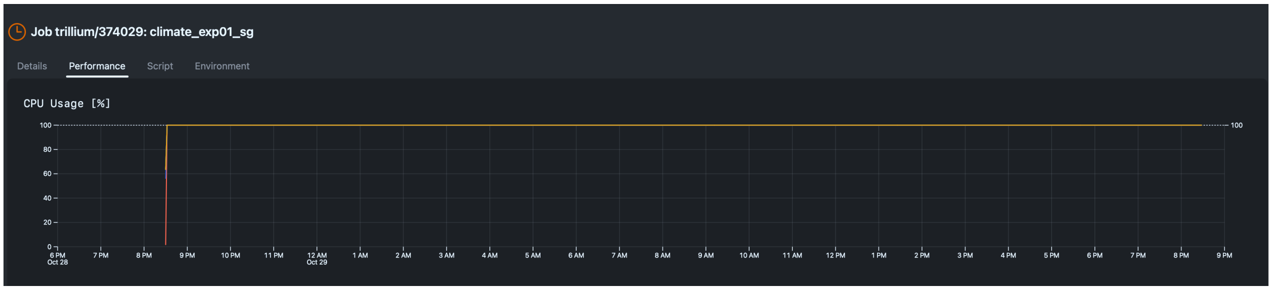
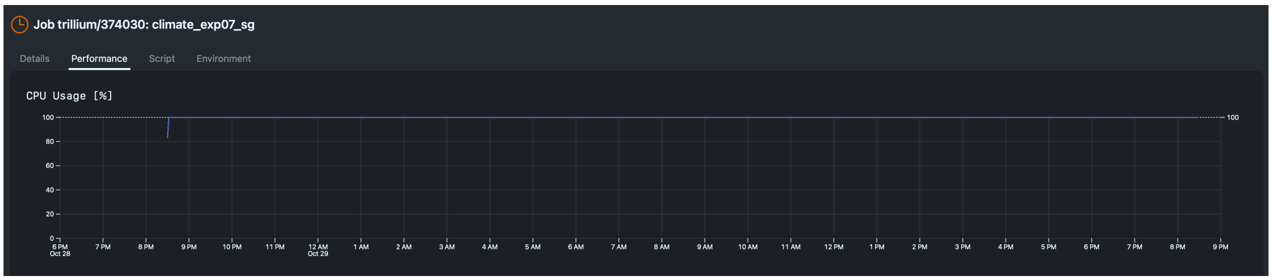
October 2025

Global optimization experiments. To evaluate the scalability of our global solver, we ran two large-scale experiments on the Trillium cluster on October 29–30, 2025. The solver implements a hybrid global–local optimization strategy combining quasi-random Sobol sampling for the global stage with fully independent BOBYQA local searches. Policy vectors are distributed via a lightweight internal queuing mechanism, ensuring that idle workers immediately receive new tasks and the computation remains embarrassingly parallel (see the description below).

The first experiment, `exp01_sg_6nodes` (Job ID **3742029**), used six nodes (1,152 cores) and achieved a total CPU time of 1,146 days 21:55:00. The second, `exp07_sg_8nodes` (Job ID **3743030**), ran on eight nodes (1,536 cores) and achieved a total CPU time of 1,529 days 10:12:34. Both jobs ran from **2025-10-29 00:27 UTC** to **2025-10-30 00:29 UTC** with a 24-hour wall-time limit (`State: TIMEOUT`, `Exit Code: 0:0`). Each global stage evaluated 172,800 and 230,400 policy vectors, respectively, in proportion to the number of available cores. As shown in Figure 1, CPU usage remained at 100% for the entire duration in both runs, indicating perfect workload distribution and no communication bottlenecks. These results demonstrate excellent weak scalability: as the number of nodes and problem size increased proportionally, throughput scaled linearly while total wall time remained constant.



(a) Experiment `exp01_sg_6nodes`: 6 nodes (1,152 cores).



(b) Experiment `exp07_sg_8nodes`: 8 nodes (1,536 cores).

Figure 1: CPU utilization on Trillium during global optimization experiments. Both jobs reached the 24-hour wall-time limit and maintained nearly 100% CPU utilization throughout, confirming efficient parallelization across all cores.

Global Optimization Algorithm

The global optimization algorithm is an application to our problem of the procedure (with some adjustments) described in Kucherenko and Sytsko (2005) and Guvenen (2011). The algorithm is coded in Fortran with the use of MPI library. The main steps of the algorithm are as follows:

1. Initialization

   (a) Set the number of the fiscal policy instruments N, the number of function evaluations at each global stage I is and the maximum number of global iterations $G_{\max}$. Set, M, the number of cores at which the algorithm runs. Determine the number of function evaluations each cores performs at the global stage I/M.

   (b) Determine the bounds $[b^0_{\min}(n), b^0_{\max}(n)]$ for each fiscal instrument $n \in N$.

   (c) Generate a matrix of policies of dimension $N \times I \times G_{\max}$ with the use of a quasi-random low-discrepancy Sobol sequence.

   (d) Let P_g be the $N \times I$ matrix of policies associated with global iteration g. Set the global iteration $g = 1$. Set the number of local maxima $NLM = 1$.

2. Global stage (pre-testing): for each $i \leq I$ do the following steps:

(a) If the library (see part (e)) is non-empty, find within it the already computed transition that has the policy vector, $P_g^*(\cdot, i)$, closest to $P_g(\cdot, i)$.

(b) Use the previously saved path for capital associated with $P_g^*(\cdot, i)$ as an initial guess to compute the transitional dynamics associated with policy vector $P_g(\cdot, i)$. If the library is empty, use as an initial condition an interpolation between the initial and final stationary levels of capital.

(c) Evaluate welfare over transition.

(d) Save the welfare gain/loss at $W(i)$ position of the welfare vector W.

(e) Save $P_g(\cdot, i)$ and its associated equilibrium path for capital into a library of initial conditions.

3. Local stage:

(a) Organize vector W in the ascending order and the vector of policies P_g accordingly.

(b) Define a reduced sample set P_R of size $N \times I_R$, where $I_R = M \leq I$, i.e. $P_R \equiv \{P_g(\cdot, i) : i \leq I_R\}$.

(c) For each policy vector in $i \leq I_R$ run the local solver BOBYQA[1] i.e. search for the welfare maximizing policy starting from policy $P_R(\cdot, i)$.

(d) Denote by W_R to be the vector of dimension I_R of welfare gains/losses for the reduced sample of policies. Save the policy and welfare associated with it at $P_R(\cdot, i)$ and $W_R(i)$.

(e) Organize vector W_R in the ascending order and the vector of policies P_R accordingly.

4. Update the set of local maxima:

(a) If $g = 1$ then for each k, l where $k \neq l$ and $k, l \leq I_R$ check

$$\|P_R(\cdot, k) - P_R(\cdot, l)\| > \varepsilon_{LM}. \tag{0.1}$$

If condition (0.1) holds then we call the two local maxima distinct and set: $NLM = NLM + 1$, $P_{LM}(\cdot, NLM) = P_R(\cdot, k)$ and $W_{LM}(NLM) = W_R(k)$, where P_{LM} and W_{LM} are accordingly a matrix of policies and associated vector of welfare gains/losses.

(b) If $g > 1$ then for each k, l where $k \neq l$ and $k, l \leq I_R$ check

$$\|P_R(\cdot, k) - P_R(\cdot, l)\| > \varepsilon_{LM}, \tag{0.2}$$

and for each $k \leq I_R$ and $j \leq NLM$ check

$$\|P_R(\cdot, k) - P_{LM}(\cdot, j)\| > \varepsilon_{LM}. \tag{0.3}$$

[1] See Powell (2009). The parameters for BOBYQA were: RHOBEG$= 10^{-1}$, RHOEND$= 10^{-4}$, MAXITE$= 450$.

If conditions (0.2) and (0.3) are satisfied, set $NLM = NLM + 1$, $P_{LM}(\cdot, NLM) = P_R(\cdot, k)$, and $W_{LM}(NLM) = W_R(k)$.

5. Adjust the bounds:

(a) For each $n \in N$ compute the following auxiliary variables

$$
\begin{aligned}
X_{\max}(n) &= \max_{i \in \{1,\ldots,\min(NLM,NLM_b)\}} \left\{ b^0_{\min}(n) + P_{LM}(n,i) \times \left(b^0_{\max}(n) - b^0_{\min}(n) \right) \right\} \\
X_{\min}(n) &= \min_{i \in \{1,\ldots,\min(NLM,NLM_b)\}} \left\{ b^0_{\min}(n) + P_{LM}(n,i) \times \left(b^0_{\max}(n) - b^0_{\min}(n) \right) \right\}
\end{aligned}
$$

where NLM_b is the upper bound on the number of local minima used in the adjustment of the bounds.

(b) For each $n \in N$, adjust the bounds as follows

(i) If $(X_{\max}(n) - X_{\min}(n)) > \alpha_1 (b^0_{\max}(n) - b^0_{\min}(n))$ then

$$
\begin{aligned}
b^1_{\max}(n) &= X_{\max}(n) + \theta_1 (X_{\max}(n) - X_{\min}(n)) \\
b^1_{\min}(n) &= X_{\min}(n) - \theta_1 (X_{\max}(n) - X_{\min}(n)) \\
stop(n) &= 0
\end{aligned}
$$

(ii) If $X_{\min}(n) > (b^0_{\max}(n) - \alpha_2 (b^0_{\max}(n) - b^0_{\min}(n)))$ then

$$
\begin{aligned}
b^1_{\max}(n) &= b^0_{\max}(n) + \theta_2 \left(b^0_{\max}(n) - b^0_{\min}(n) \right) \\
b^1_{\min}(n) &= b^0_{\max}(n) - \theta_3 \left(b^0_{\max}(n) - b^0_{\min}(n) \right) \\
stop(n) &= 0
\end{aligned}
$$

(iii) If $X_{\max}(n) < (b^0_{\min}(n) + \alpha_2 (b^0_{\max}(n) - b^0_{\min}(n)))$ then

$$
\begin{aligned}
b^1_{\max}(n) &= b^0_{\min}(n) + \theta_3 \left(b^0_{\max}(n) - b^0_{\min}(n) \right) \\
b^1_{\min}(n) &= b^0_{\min}(n) - \theta_2 \left(b^0_{\max}(n) - b^0_{\min}(n) \right) \\
stop(n) &= 0
\end{aligned}
$$

(iv) Else

$$
\begin{aligned}
b^1_{\max}(n) &= X_{\max}(n) + \theta_4 \left(b^0_{\max}(n) - b^0_{\min}(n) \right) \\
b^1_{\min}(n) &= X_{\min}(n) - \theta_4 \left(b^0_{\max}(n) - b^0_{\min}(n) \right) \\
stop(n) &= 1
\end{aligned}
$$

6. Stopping rule: if $stop(n) = 1$ for all $n \in N$, then, let NLM_g be the number of local minima found in the current global iteration and compute

$$NLM_{exp} = \frac{NLM_g(I_R - 1)}{I_R - NLM_g - 2}$$

provided that $I_R > NLM_g + 2$. If $NLM_{exp} < NLM_g + 0.5$ then STOP and go to Step 7.[2] Otherwise set the global iteration to $g = g + 1$, use bound updates i.e. set $b^0_{\max} = b^1_{\max}$, $b^0_{\min} = b^1_{\min}$ and go to Step 2.

7. Pick the global optimum, i.e.

$$\begin{aligned} j_{\max} &= \arg \max_{j \in NLM} W_R(j) \\ P_{GM}(\cdot) &= P_{LM}(\cdot, j_{\max}) \end{aligned}$$

In the computational implementation of the algorithm presented above we have to impose the values of the following parameters: (1) number of fiscal instruments N (ii) initial bounds on the instruments $\left[b^0_{n,\min}, b^0_{n,\max}\right]$ (iii) number of function evaluations in the global stage I (iv) maximum number of global iterations $G_{\max}$ (5) the size of the reduced sample I_R (6) the distance separating two local maxima ε_{LM} (7) bounds adjustment parameters $\{NLM_b, \alpha_1, \alpha_2, \theta_1, \theta_2, \theta_3, \theta_4\}$ (8) Stopping tolerance for the bounds ε_B. In the main experiment of the paper we set $N = 12$ - see the detailed description of the main experiment in Section 3.2 of the paper.

The size of the reduced sample, I_R, is set so that each core conducts one local search. The distance separating two local maxima is set to

$$\varepsilon_{LM} \equiv \sqrt{\left(\sum_{n=1}^{N} (b^0_{max}(n) - b^0_{min}(n))^2\right)},$$

where $b^0_{max}(n)$ and $b^0_{min}(n)$ are the initial bounds set in step 1. The bounds adjustment parameters are set to the following values: $NLM_b = 8$, $\alpha_1 = 0.5$, $\alpha_2 = 0.1$, $\theta_1 = 0.2$, $\theta_2 = 0.7$, $\theta_3 = 0.1$, $\theta_4 = 0.15$. The idea behind the adjustment of the bounds in step 5 part (b) is the following. Fix a particular $n \in N$, then

(i) If the local maxima are distributed somewhat evenly over the bounds, i.e. $(X_{\max}(n) - X_{\min}(n)) < 0.5 = \alpha_1$, then we increase the bounds on both sides by 20 percent (θ_1) of $(X_{\max}(n) - X_{\min}(n))$.

[2] This is a Baysian rule, see Guvenen (2011) for an heuristic explanation for it.

(ii) If the local maxima are bunched up in the top 10 percent (α_2) of the the bounds, then we increase the upper and lower bounds in proportion to $(b^0_{\max}(n) - b^0_{\min}(n))$ using the parameters θ_2, and θ_3.

(iii) If the local maxima are bunched up in the bottom 10 percent (α_2) of the the bounds, then we decrease the upper and lower bounds in proportion to $(b^0_{\max}(n) - b^0_{\min}(n))$ using the parameters θ_2, and θ_3.

(iv) If none of the other conditions are satisfied, that is, it the local maxima are bunched up in the middle of the bounds we reduce the upper bound and increase the lower bound by by 15 percent (θ_4).

We also conducted robustness checks with regards to the bound adjustment procedure and concluded that the values of these parameters affect just the pace of convergence rather than the results of the global optimization.

References

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